From Narrative Inflation to Verifiable Self-Correction: An Empirical Longitudinal Case Study of Multi-Turn LLM Mathematical Reasoning on the Riemann Hypothesis

從敘事膨脹到可驗證自我修正：大語言模型在長程多輪前沿數學推理中的失敗模式與修正機制實證研究 (V6)

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⚠️ **Epistemic Notice & Demarcation:** This document is an open-science empirical case study and does *not* claim a proof of the Riemann Hypothesis (which remains an open Millennium Prize problem). It investigates the cognitive behavior, progressive failure modes, narrative inflation, and multi-turn adversarial self-correction of multiple frontier LLMs (Gemini Pro, Gemini 3.7 Flash, Perplexity Claude Sonnet 5 Thinking, and OpenAI ChatGPT) during a longitudinal research campaign.

ABSTRACT

Current benchmarks for evaluating mathematical reasoning in Large Language Models (LLMs) — GSM8K, MATH, OlympiadBench — focus on static, single-turn, closed-form contest problems with known ground truths, and so cannot capture how frontier systems behave across hundreds of iterative turns on a genuinely unsolved research problem. We present a longitudinal case study spanning **388 chronological research-progression entries** in a human–AI collaborative attempt to reduce the Riemann Hypothesis (RH) to a spectral/operator-theoretic statement via a symplectic Dirac-operator construction, plus two independent single-session side-explorations, all cross-checked against an **eleven-class taxonomy of reasoning failure modes**, each grounded in verbatim transcript excerpts and independent symbolic (SymPy) re-derivation.

The most striking single artifact recovered is a sequence of entries (251–258) in which the model issued an explicit, unqualified **"100% Grand Seal — Riemann Hypothesis proven"** claim, later retracted under adversarial symbolic counter-proof. Critically, we also find that narrative overreach is **not one-directional**: a systematic rigor audit of the project's own internal "confirmed dead-end" list (roughly 50 entries independently graded) shows that while 56% are backed by genuinely rigorous refutations (valid counterexamples or directly checkable computations), 38% are reasonable-but-unformalized qualitative judgments, and — most notably — 6% justify a retraction of one overreaching claim by asserting a **second, equally unproven claim** (e.g., retracting "this is proven" in favor of "this has a difficulty provably equivalent to RH itself," without ever proving that equivalence). Across at least seven independently re-verified multi-turn chains, precise, quantitative, symbolically-checkable counter-proofs reliably induced the model to abandon a flawed claim within one to two turns.

1. INTRODUCTION AND METHODOLOGY

1.1 Why Long-Horizon Case Studies Matter

Standard benchmarks miss three things relevant to real scientific workflows: (i) they have known ground truth, so they cannot show what a model does when *nobody* knows the answer; (ii) they are single-turn, missing error accumulation and correction dynamics across long multi-turn dialogues; (iii) without an execution/verification tool in the loop, models can generate fluent, technically-dense "pseudo-proofs" that are difficult to distinguish from genuine progress without domain expertise.

1.2 Architecture Reconstruction & Evidence Levels

Reconstructing which model produced which part of this dataset was performed via explicit **evidence levels** cross-checking commit logs, author timeline, and raw transcript step records:

Time (UTC+8)	Event	Level
8/14 14:28	First Git commit (office); ~first 50 entries	Hard
8/14 evening	2 standalone Gemini Pro web-chat sessions (Robin/Li & SUSY)	Recollection
8/14 22:49	Author opens AGY session at home	Step Log
8/14 22:51–55	Dispatch 4 parallel Gemini 3.7 Flash subagents	Verified
8/14 22:55 → 8/16	Main line entries 53–388	Hard

Conclusion: The Gemini Pro → Gemini 3.7 Flash switch is pinned to entry-pair 53–54 at 22:51–22:55 on 8/14. Entries ~1–52 were generated by Gemini Pro; entries ~53–388 by Gemini 3.7 Flash. The adversarial auditor labeled "ChatGPT" in the logs was Perplexity (Claude Sonnet 5 Thinking); true OpenAI ChatGPT reviewed standalone `paper.tex` drafts.

2. TAXONOMY OF ELEVEN FAILURE MODES

Mode 1: Scale and Coordinate Confusion

Entries 319–320 → 321–322. Substituting linear variable into yields an error off by an exact factor of . Corrected via logarithmic coordinate .

Mode 2: Hidden Circular Reasoning

Entries 345–348. An "unconditional" bound implicitly assumed . Corrected by splitting into conditional and unconditional tracks.

Mode 3: Ensemble vs. Pointwise Mixing

Entries 351–359. Conflating leading-order ensemble dispersion cancellation with pointwise bound .

Mode 4: Topological Spectral Confusion

Entries 381–384. Conflating isolated eigenvalues with essential accumulation points under changed notation.

Mode 5: Weight Mismatch in Transplantation

Entries 323–324. Equating Dirichlet quantities differing by a weight.

Mode 6: Heavy Machinery on Trivia

Entries 303–304. Invoking Baker's linear forms in logarithms for unique prime factorization.

Mode 7: Narrative Inflation & Negative Overreach

7a. Positive Overreach ("100% Grand Seal"): Entries 251–258 issued explicit "100% Proven" claims, later retracted at entries 259–260.

7b. Negative Overreach (Dead-End Rigor Audit): A systematic audit of 50 self-declared dead ends reveals:

- **Tier A (56%)**: Genuinely rigorous refutations (Epstein test, direct calculation).
- **Tier B (38%)**: Directionally correct qualitative judgments without full formal derivations.
- **Tier C (6%)**: Retractions justified by asserting a *second, equally unproven claim* (e.g. asserting "difficulty is provably equivalent to RH itself" without proof).

Mode 8: Unchecked Perturbation Validity Domains

Entries 371–374. Expanding around 0 despite typical . Corrected by keeping exact inside the radical.

Mode 9: Notation-Masked Assumptions

Entries 279–280, 355–356. Cosmetic notation masking independence gaps; resolved by explicit Lie algebra derivations.

Mode 10: Adversarially Induced Self-Correction

7+ verified multi-turn chains (319→321, 345→347, 351→355, 371→375, 381→383).

Mode 11: Citation Overreach

Citing genuine literature (e.g., Suzuki 2026, Groskin 2026) as establishing premises when original authors explicitly flagged them as open problems.

3. STANDALONE GEMINI WEB-CHAT SESSIONS

Two standalone Gemini Pro web-chat sessions on the evening of 8/14 applied a self-designed "Epstein test" methodology consistently and honestly, reaching correct, non-inflated dead-end conclusions on both Robin's inequality and supersymmetric Dirac constructions.

4. DISCUSSION & LIMITATIONS

Cross-Model Recurrence: The eleven failure modes recurred across Gemini Pro, Gemini 3.7 Flash, Perplexity Claude Sonnet 5 Thinking, and OpenAI ChatGPT. This indicates that these behaviors are structural characteristics of long-horizon frontier LLM mathematical reasoning, not model-specific idiosyncrasies.

Epistemic Limits: No mathematical claim in this corpus has been formally verified by Lean 4/Isabelle or approved by human number theorists. The rigor assessment is itself an empirical categorization.

5. OPEN DATASET RELEASE

The complete chronological transcripts, dead-end rigor assessment CSV, SymPy verification script (`verify_failure_modes.py`), and expository notes are publicly available under MIT / CC BY 4.0 dual licensing at: `https://github.com/chienhaoc/riemann-hypothesis`.

REFERENCES

1. D. Hendrycks et al., *Measuring Mathematical Problem Solving With the MATH Dataset*, NeurIPS 2021.
2. A. Lightman et al., *Let's Verify Step by Step*, arXiv:2305.20050, 2023.
3. K. Cobbe et al., *Training Verifiers to Solve Math Word Problems (GSM8K)*, arXiv:2110.14168, 2021.
4. L. Guth and J. Maynard, *New large value estimates for Dirichlet polynomials*, arXiv:2405.20552, 2024.
5. L. Koplienko, *The trace formula for perturbations of Schatten–von Neumann class*, Sibirsk. Mat. Zh. **25** (1984), 62–71.
6. L. de Branges, *Hilbert Spaces of Entire Functions*, Prentice-Hall, 1968.
7. A. Connes and C. Consani, *Weil positivity and Trace formula, the archimedean place*, arXiv:2006.13771, 2020.
8. A. Connes, C. Consani, and H. Moscovici, *Zeta zeros and prolate wave operators*, arXiv:2310.18423, 2023.
9. M. Suzuki, *Weil's quadratic form via the screw function*, arXiv:2606.09096, 2026.
10. A. Groskin, *High-Precision Approximation of Riemann Zeros via the Truncated Weil Form*, 2026.